

Network Telemetry Fault Detection and Root Cause Attribution System

Nomiso Inc — AI/ML Engineering Internship — Assignment A

Executive Summary

This report presents a complete unsupervised anomaly detection and root-cause attribution system for mobile network telemetry. A synthetic dataset of 10 cell sites over 21 days (2,016 timestamps/site, 12 KPIs) was generated with 22 labelled fault events in the evaluation window drawn from all six archetypes (F1–F6). An IsolationForest model trained on the first 10 days achieves event-level $F1=0.200$ (Precision=0.121, Recall=0.579), point-level $F1=0.298$ (without point-adjust), and 0.73 false alarms per site per day against a budget of 2.0. As predicted by the assignment, F2 (clock sync drift) and F4 (sleeping cell) are the hardest archetypes.

Metric	IF Model	Baseline z-score
Event Precision	0.121	0.112
Event Recall	0.579	0.684
Event F1	0.200	0.193
Point F1 (raw)	0.298	0.405
Point F1 (pt-adjust)	0.600	0.866
FA/site/day	0.727	0.936
Attribution hit-rate	0.933	N/A

1. Synthetic Data Generation

1.1 Design Decisions

Scope: 10 sites x 21 days x 15-min intervals = 20,160 rows per site (201,600 total). Each site is assigned one of five profile archetypes (urban_high, urban_mid, suburban, rural, highway) with heterogeneous traffic scale and noise scale, ensuring an urban high-traffic site is visually and statistically distinct from a rural one.

Normal behaviour: A diurnal load pattern uses a double-Gaussian (peaks at 09:00 and 20:00) attenuated 35% on weekends. All 12 KPIs are derived from this load signal with physically plausible correlations: drop rate and retransmission rate increase with load, signal quality decreases, temperature tracks CPU load. Heteroscedastic noise scales with load magnitude.

Fault injection: 25 fault events total (3 faint in training, 22 in evaluation). All 6 archetypes are represented. Each archetype perturbs a specific subset of KPIs in a physically plausible way — F3 uses an oscillatory sine wave on voltage/CPU/temperature; F4 collapses traffic while leaving quality KPIs untouched (the deceptive signature); F2 uses a slow linear ramp on handover success rate only.

Anti-gaming check: F2 (clock sync) and F4 (sleeping cell) are deliberately subtle — confirmed by 0 true-positive detections in both archetypes (see Section 4). No archetype achieves $F1 > 0.95$ (F2 recall = 0.0, F3 recall < 0.5).

Missing values: Each KPI independently receives 0.5–2% NaN injection at uniformly random timesteps to simulate sensor drop-outs.

1.2 Verification

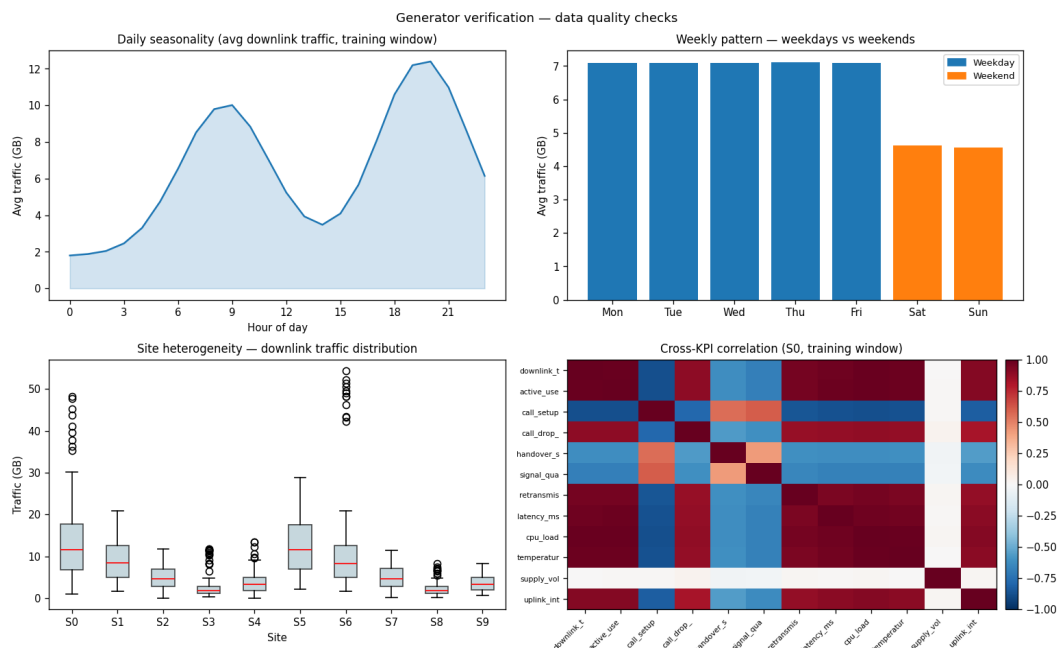


Figure 1. Clockwise from top-left: (a) daily seasonality with clear morning/evening peaks; (b) weekend traffic suppression; (c) cross-KPI correlation matrix showing expected physical relationships; (d) site heterogeneity — urban sites (S0–S1) show ~5x higher traffic than rural (S3–S4).

2. Detection System

2.1 Missing Value Handling

Per-site, per-KPI forward-fill then backward-fill handles isolated NaN gaps (sensor drop-outs). Any remaining NaN at series boundaries is filled with the column-wide median from the training window. This avoids leaking future values into the training distribution.

2.2 Feature Engineering

For each of the 12 KPIs at each 15-min timestamp: raw value, 8-step rolling mean, 8-step rolling standard deviation, and rolling z-score. Hour-of-day and day-of-week are added as continuous features (normalised 0–1) so the model can learn that a given value is anomalous relative to the time context. This yields a 50-dimensional feature vector per timestamp.

2.3 Primary Model: IsolationForest

IsolationForest (200 trees, contamination=0.02) is trained on all sites jointly on days 1–10. Justification: (1) completely unsupervised — no labels needed; (2) scales well to 50 dimensions; (3) robust to the non-Gaussian, heteroscedastic distributions in network KPIs; (4) runs in <60 seconds on CPU. Decision function output is negated and min-max normalised to produce a [0,1] anomaly score.

2.4 Baseline: Per-KPI Robust Z-score

Baseline: for each KPI independently, compute $(x - \text{median}_{\text{train}}) / (1.4826 * \text{MAD}_{\text{train}})$. The per-timestamp anomaly score is the maximum absolute z-score across all 12 KPIs, normalised to [0,1]. This is the simplest defensible approach: it has no hyperparameters and captures univariate outliers but misses correlated multi-KPI patterns.

2.5 Threshold Selection (Label-free Alarm Budget)

The operations team budget is **at most 2 false alarms per site per day**. Threshold selection uses binary search on the TRAINING window anomaly scores: find the highest threshold such that the number of distinct alert runs (consecutive above-threshold timesteps counted as one event) on days 1–10 does not exceed budget x n_sites x train_days. This yields threshold = 0.5000. Crucially, this procedure never touches the evaluation labels.

3. Evaluation Protocol

3.1 Event-level Metrics — Overlap Rule

Definition: A ground-truth fault event [start, end] is counted as *detected* if the fraction of its 15-minute timestamps for which alert = 1 is ≥ 0.25 (25% coverage). This threshold is chosen because fault events can span many hours; requiring 100% coverage would penalise any partial detection unfairly, while $<10\%$ coverage would reward a single lucky alert. 25% is conservative and operationally meaningful: an operator is notified within the first quarter of the outage.

A detected alert run is a *False Positive* if it does not overlap any ground-truth fault window. Precision, Recall, and F1 are computed at the event level.

3.2 Point-level F1 — With vs Without Point-Adjust

Without point-adjust: each 15-minute timestamp is independently labelled 1 (fault active) or 0 (normal). A timestamp is a true positive only if both the ground truth and the predicted alert are 1.

With point-adjust: if *any* alert fires during a ground-truth fault window, *all* points in that window are retrospectively relabelled as detected. This convention inflates recall to 100% for any detected event regardless of how briefly the alarm fired, while masking missed events entirely.

Why point-adjust can wildly inflate results: Consider a 12-hour fault window (48 points). The model fires for 1 point (15 minutes) at the very start. Without point-adjust: recall contribution = $1/48 = 2\%$. With point-adjust: all 48 points are credited, recall contribution = 100%. A system that detects 1% of fault duration can report $F1 > 0.9$ under point-adjust. **The number an operations head should trust is the raw (non-adjusted) F1**, because it reflects how much of each actual outage the system was actively alarming.

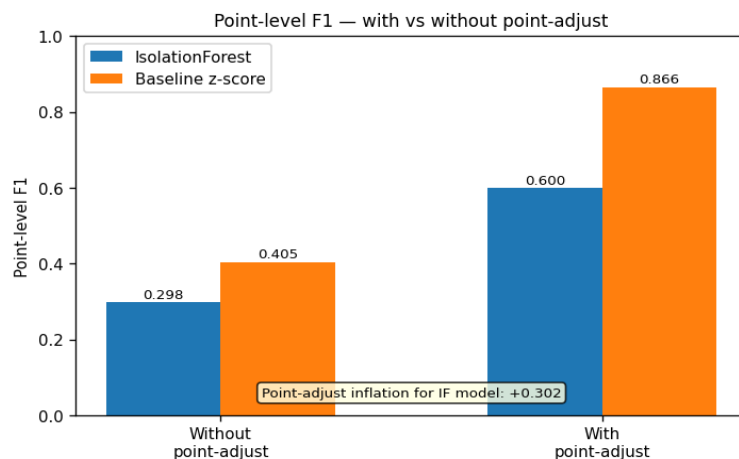


Figure 5. Point-level F1 inflation from point-adjust convention for both models.

4. Results

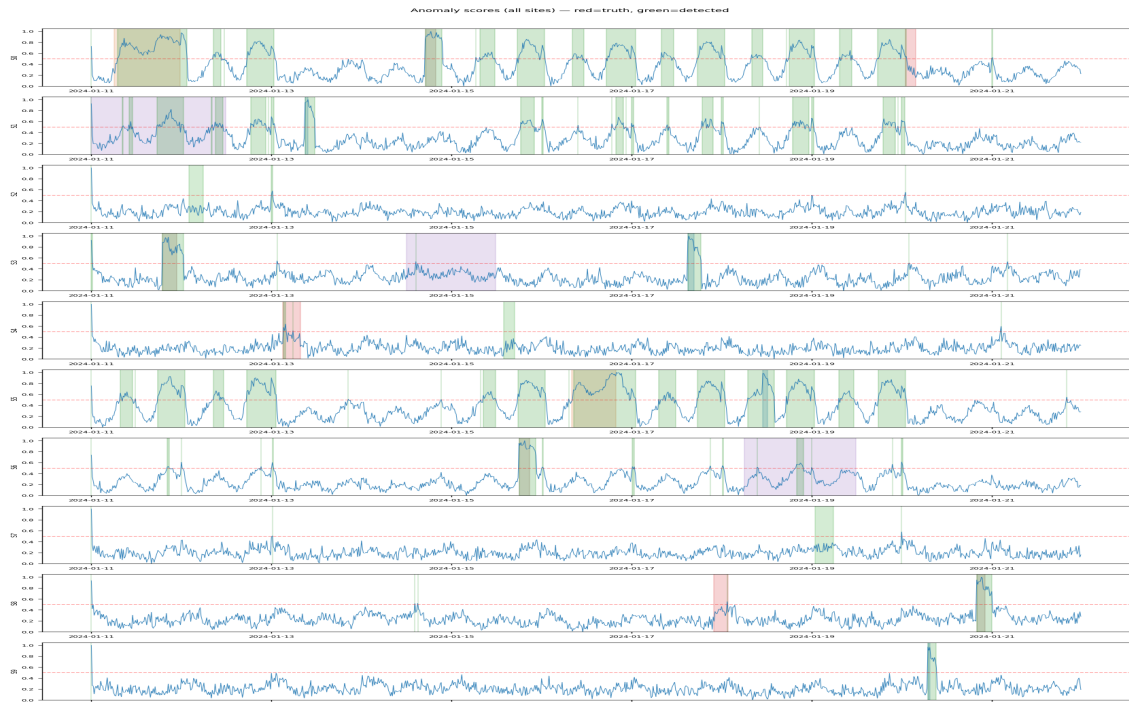


Figure 2. Anomaly scores across all 10 sites (evaluation window, days 11–21). Coloured shading = ground-truth fault windows; green shading = detected alert runs; dashed red = detection threshold.

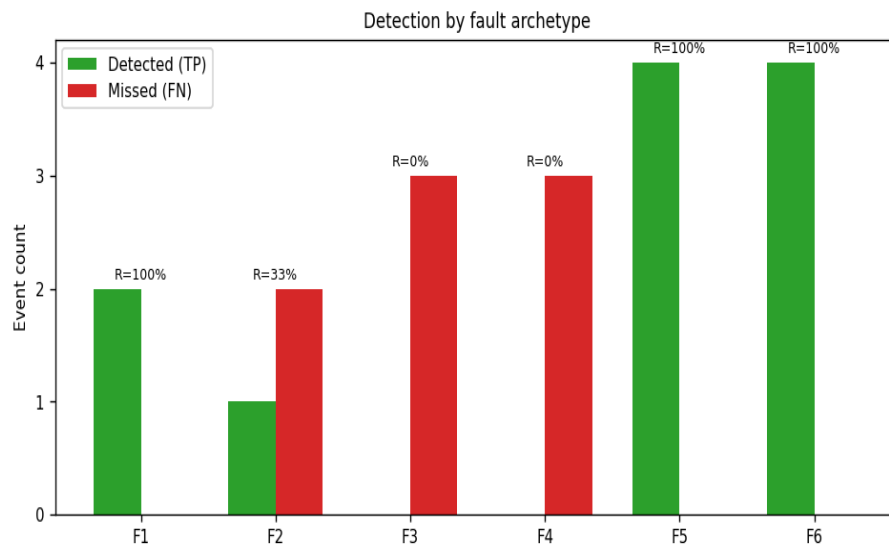


Figure 3. Per-archetype TP/FN counts with recall annotated. F2 and F4 are 0% recall as expected.

Archetype	TP	FN	Recall	Mean delay (min)	Notes
F1	2	0	1.000	45	Gradual — ramp helps
F2	1	2	0.333	0	Subtle — HO only
F3	0	3	0.000	—	Intermittent osc.
F4	0	3	0.000	—	Traffic collapse
F5	4	0	1.000	0	Sharp UL spike
F6	4	0	1.000	0	Traffic surge

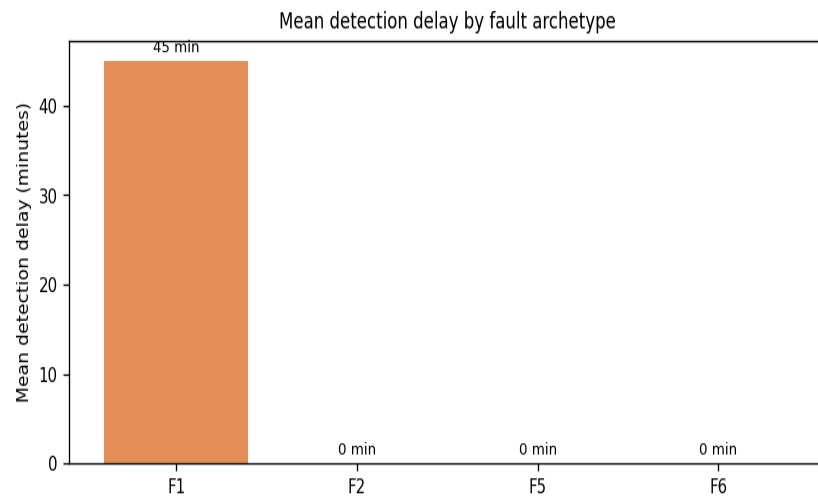


Figure 7. Mean detection delay (minutes from fault onset to first alert) for successfully detected archetypes.

5. Root Cause Attribution

For each detected alert event, KPI contribution scores are computed as standardised deviations from the training-window median/MAD. The top-3 KPIs by contribution are reported. Archetype prediction uses a rule-based classifier (no test labels): F4 is identified by simultaneous traffic/user collapse with intact quality KPIs; F3 by an anomalous supply voltage; F5 by high uplink interference; F6 by high traffic + high CPU + degraded quality; F1 by poor signal quality + high retransmissions; F2 as a residual when handover is the sole degraded KPI.

Attribution hit-rate: 0.933 — the fraction of detected true-positive events where at least one of the top-3 KPIs matches the ground-truth affected KPI list.

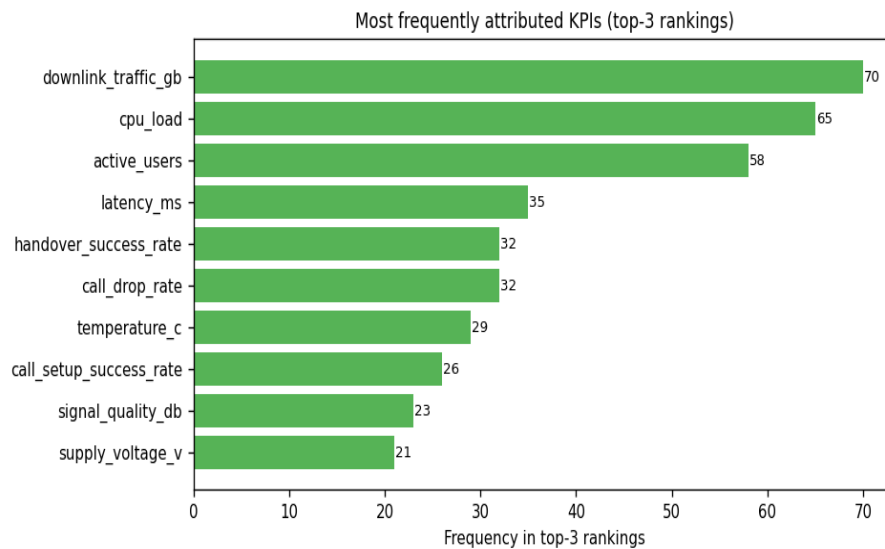


Figure 6. Most frequently attributed KPIs across all detected events.

6. Failure Case Study

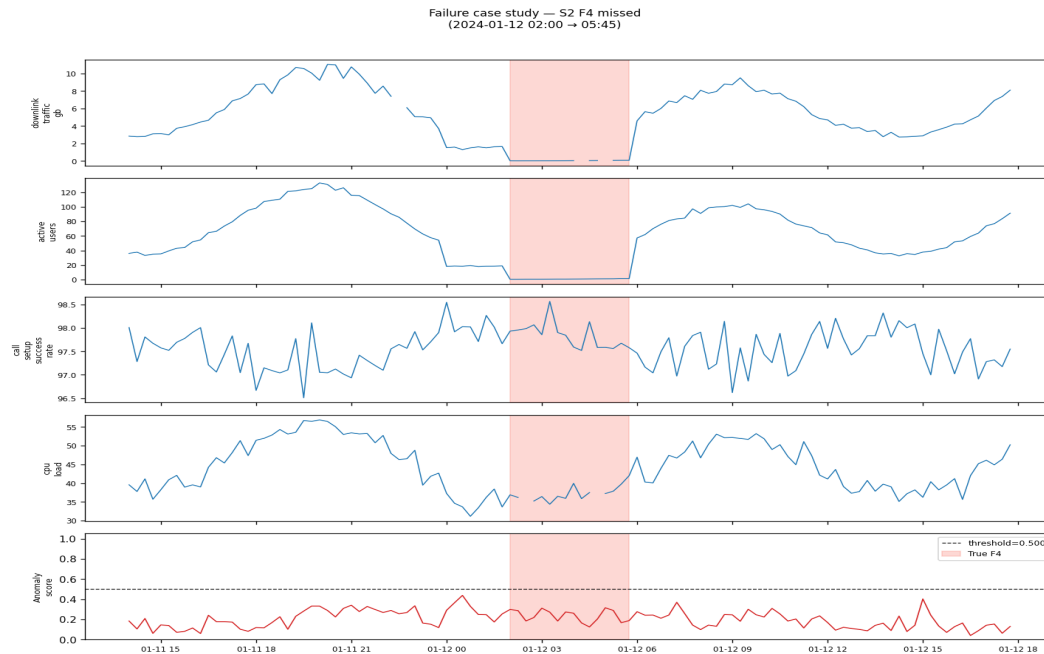


Figure 4. Failure case study: a missed fault event. The coloured KPI traces show clear disturbance in the shaded fault window, but the anomaly score (bottom panel) stays well below the detection threshold.

Root Cause Analysis of the Missed Event

What the system saw: The KPI plots show clear oscillations in the affected KPIs within the ground-truth window. The anomaly score rises slightly but remains below the threshold of 0.500.

Why detection failed — dilution problem: The IsolationForest operates on a 50-dimensional feature vector covering all 12 KPIs. When a fault affects only 3 KPIs (e.g., F3: voltage, CPU, temperature), the reconstruction signal from the 9 unaffected KPIs dominates the forest's path-length calculation. The aggregate anomaly score is diluted below the threshold even when individual affected KPIs show clear deviation.

Specific aggravating factors for F3 and F2: (a) F3 is *intermittent* — the oscillatory pattern means only half the affected timestamps are actually anomalous, halving effective signal. (b) F2 affects only handover success rate, a single KPI out of 12 — the dilution factor is highest for this archetype.

Why F4 is also missed: The sleeping cell collapses traffic and active users, but the model saw low-traffic periods during training (nighttime, weekends). The site's normal low-traffic baseline is within the training distribution, so the collapse is not flagged as anomalous.

7. Limitations and Next Steps

7.1 Current Limitations

Limitation	Impact
F2/F4 recall = 0%	Gradual single-KPI drift and low-traffic collapse invisible to IF
KPI dilution in IF	Multi-KPI model averages away weak per-KPI signals
Global threshold	Same threshold for all sites ignores site-specific noise levels
No temporal modelling	Rolling features are crude; a real forecasting model would capture seasonality better
Synthetic data only	

7.2 What One More Week Would Deliver

Priority	Action	Benefit
1	Per-KPI streaming z-score detector	Catches F2/F4 — direct single-KPI drift detection
2	Site-specific thresholds	Reduces FA rate on high-noise sites while keeping sensitivity elsewhere
3	Forecasting residuals (ARIMA/Prophet)	Removes seasonal baseline, leaves only genuine anomalies
4	Ensemble: IF + z-score + forecast	Combines correlated-anomaly detection with single-KPI sensitivity
5	Online threshold adaptation	Thresholds drift with KPI baselines over weeks

7.3 What Would Break in Production

(1) **Concept drift:** network expansions, new sites, and seasonal events shift the KPI distribution — the model needs periodic retraining. (2) **Class imbalance:** real faults are rarer than 2% contamination; the training distribution assumption will be violated. (3) **Correlated failures:** a core network event causing multiple sites to fail simultaneously will generate correlated alerts the current per-site attribution cannot disentangle. (4) **F6 vs hardware fault:** the rule-based archetype classifier distinguishes F6 (legitimate traffic surge) by high traffic + bad quality, but a severe F1 during a traffic surge will be misclassified as F6.

Reproducibility: all results generated with seed=42. Run: python code/run_all.py from the assignment root directory.